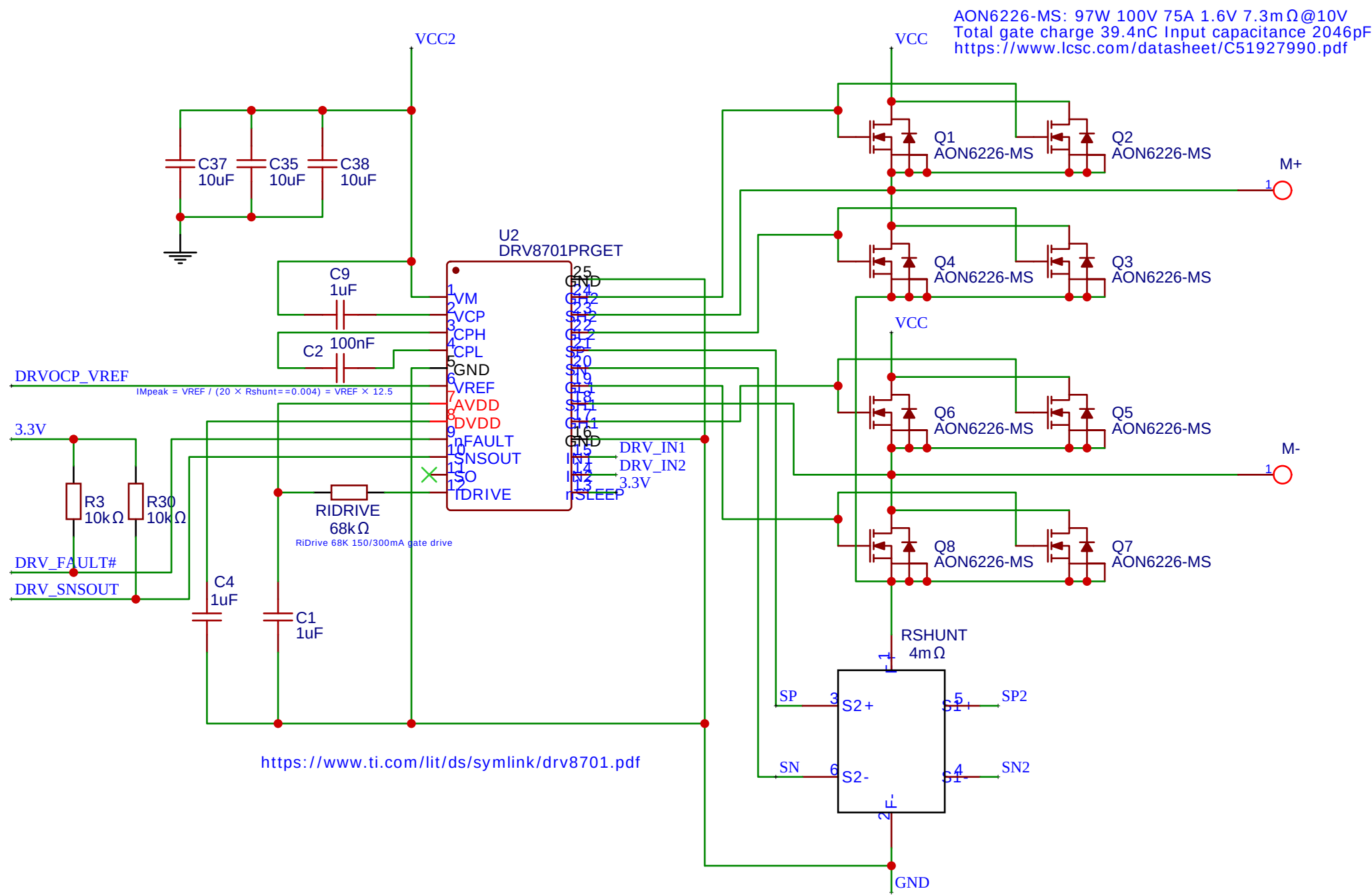


DC Motor Controller with Velocity and PWM Mode

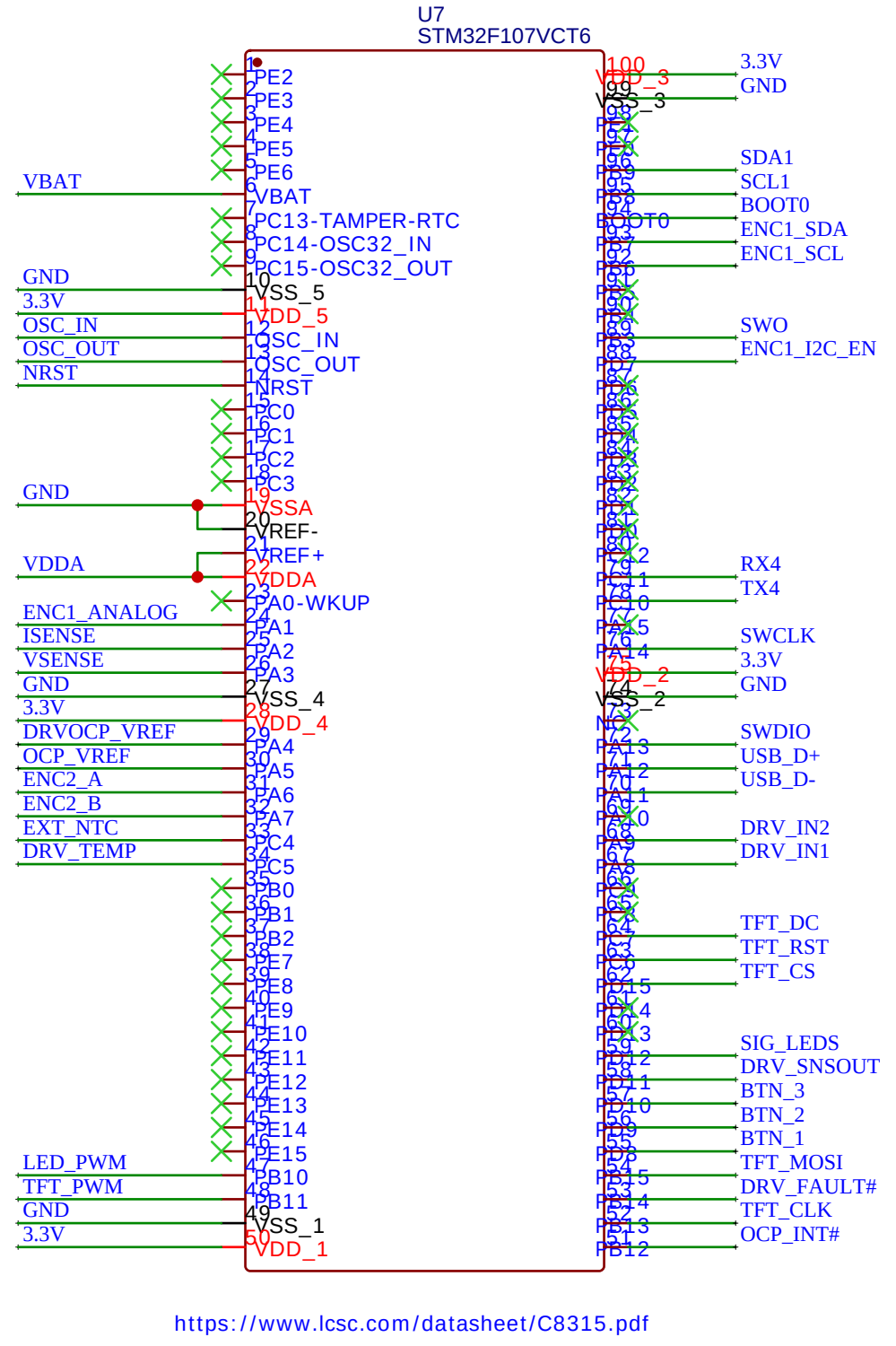
- 8-36V, 20A continuous, 40A peak
- Adjustable cycle by cycle motor current limit (0.5-40A)
- Adjustable max. current with signal LED (0.5-40A)
- 1.14" 135x240 TFT display, rotary encoder and 3 buttons for easy navigation
- Support for magnetic encoder MT6701 in A/B mode up to 1024PPR (55000 RPM)
- Dimmable LED with CC driver 3-32V/350mA/5W
- Current, voltage and temperature monitoring

Firmware: https://github.com/sascha432/motor_controller_stm32

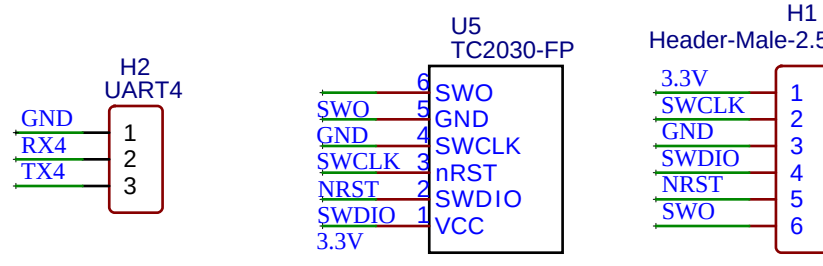
Motor driver and MOSFETS



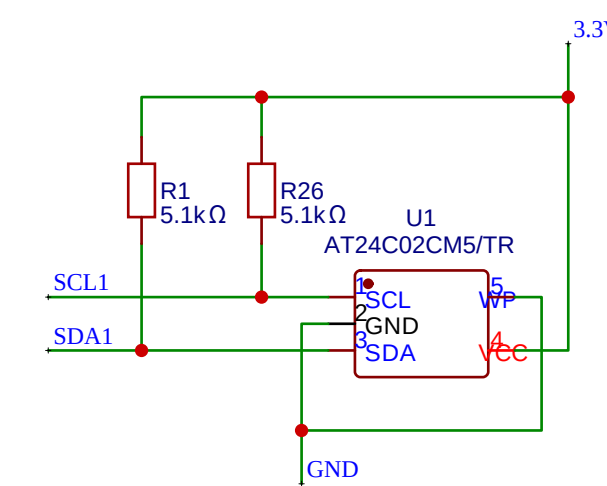
MCU and EEPROM



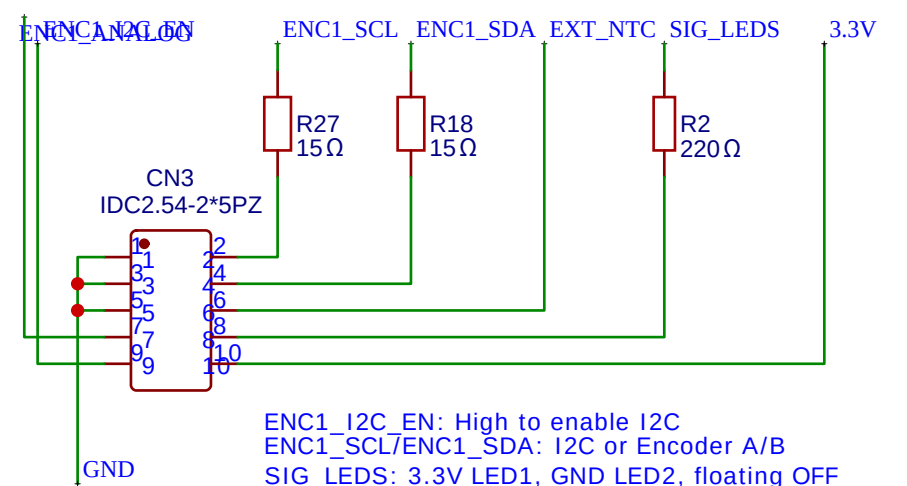
Programming/debug header



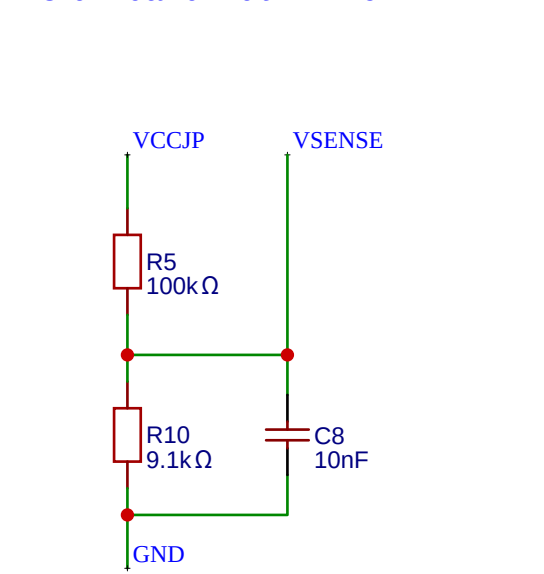
EEPROM and I2C pullups



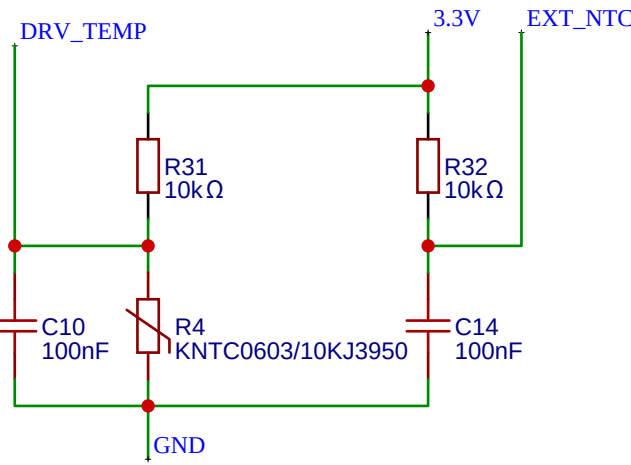
Externals & MT6701 connector



Input voltage detection up to 40V

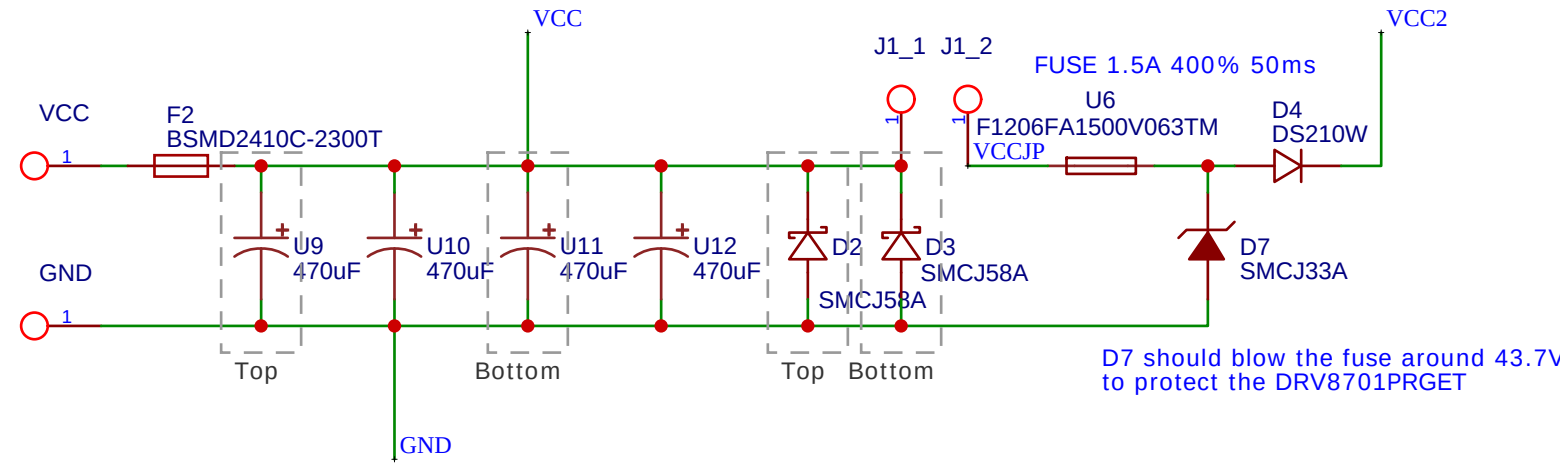


Temperature monitoring

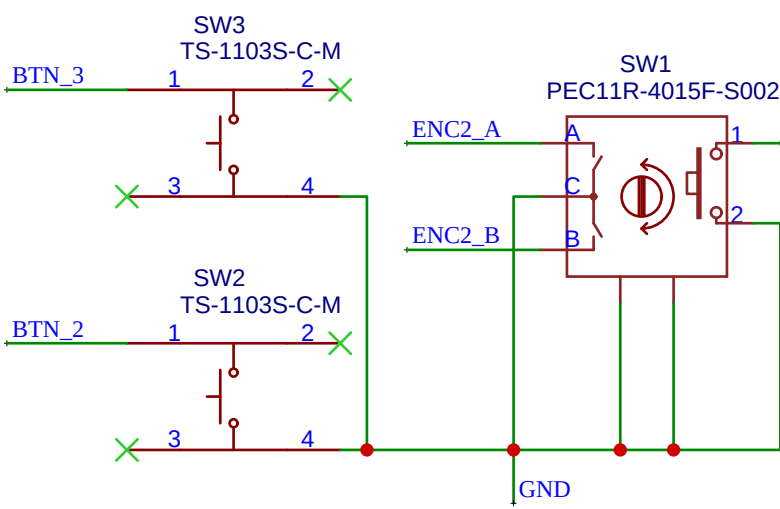


Input bulk capacitance and protection

VCC2 is decoupled from bulk capacitance and fused

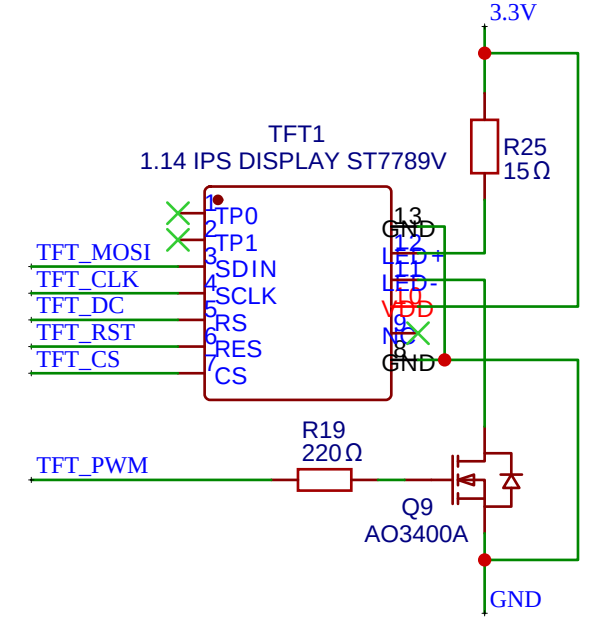


Buttons and rotary encoder

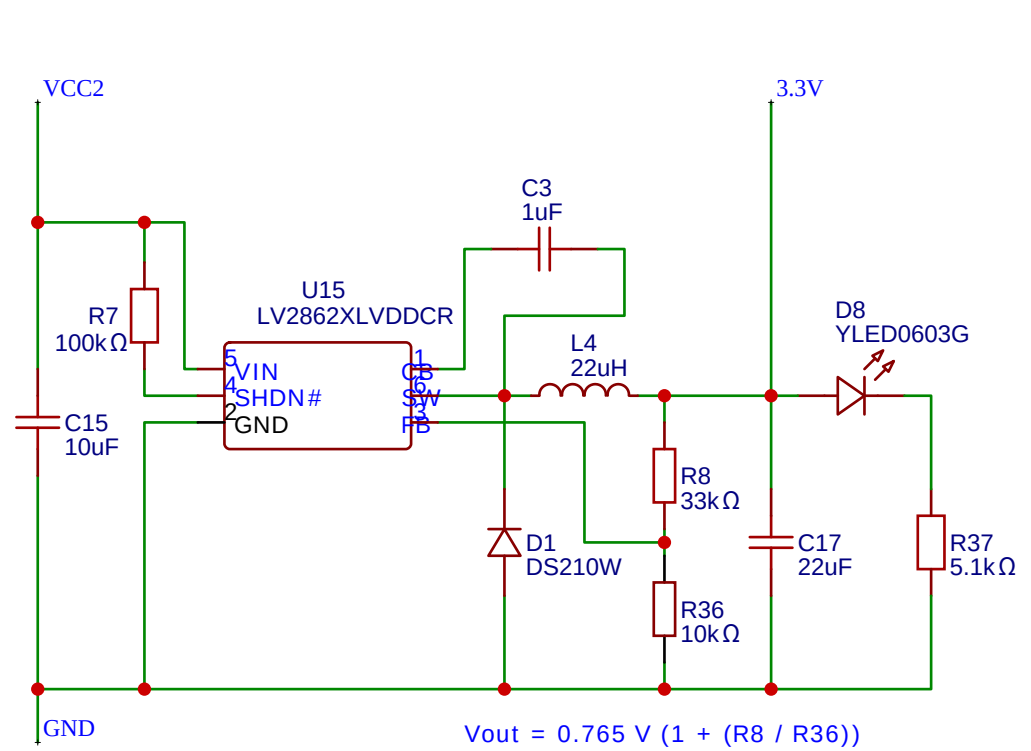


TFT

1.14" 135x240 Full Color IPS Screen 135*240 ST7789 SPI 13 Pins

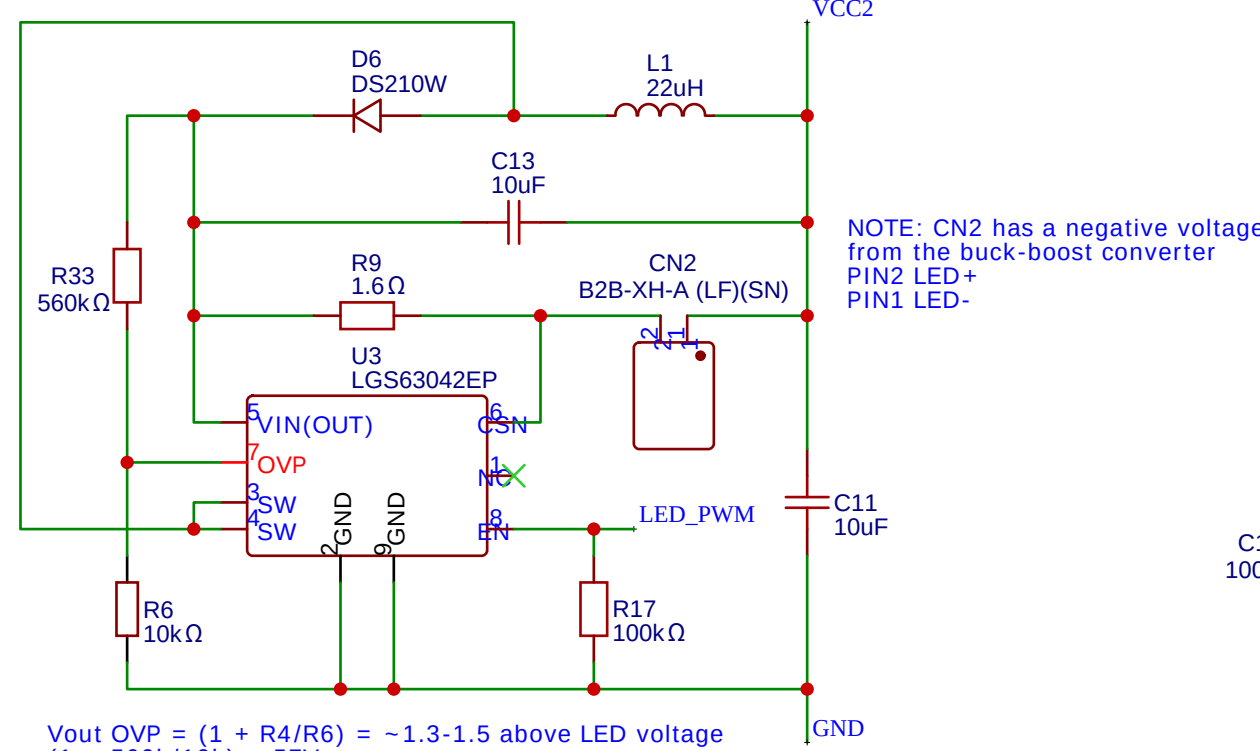


Power Supply 3.3V/0.6A



CC LED driver with PWM 2.5-32V/350mA

NOTE: If the LED voltage + VCC exceeds 57V, the OVP is triggered
(OVP - VCC) / 1.3 = max. VLED, (57V - 36V) / 1.3 = ~16V



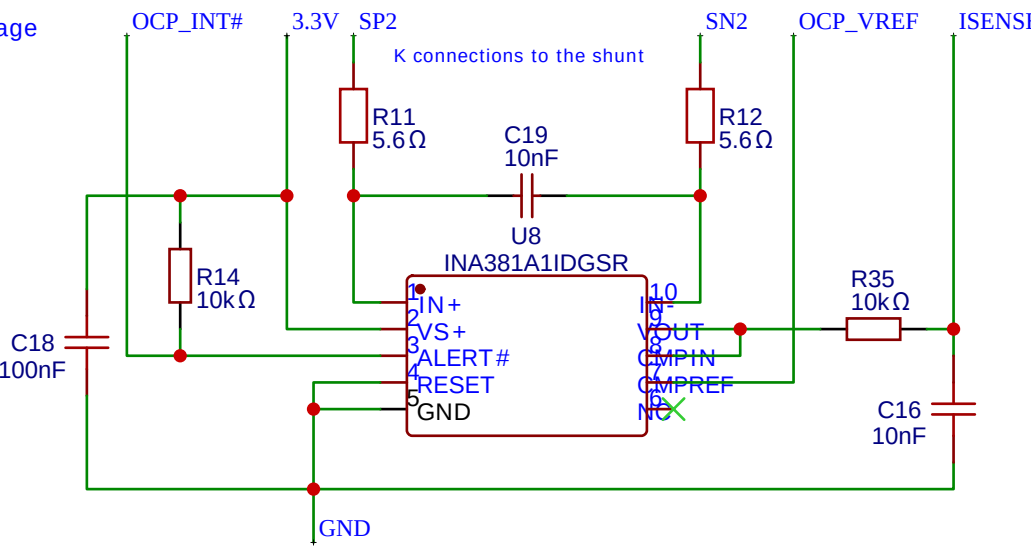
Input OCP

20x current amplifier with comparator

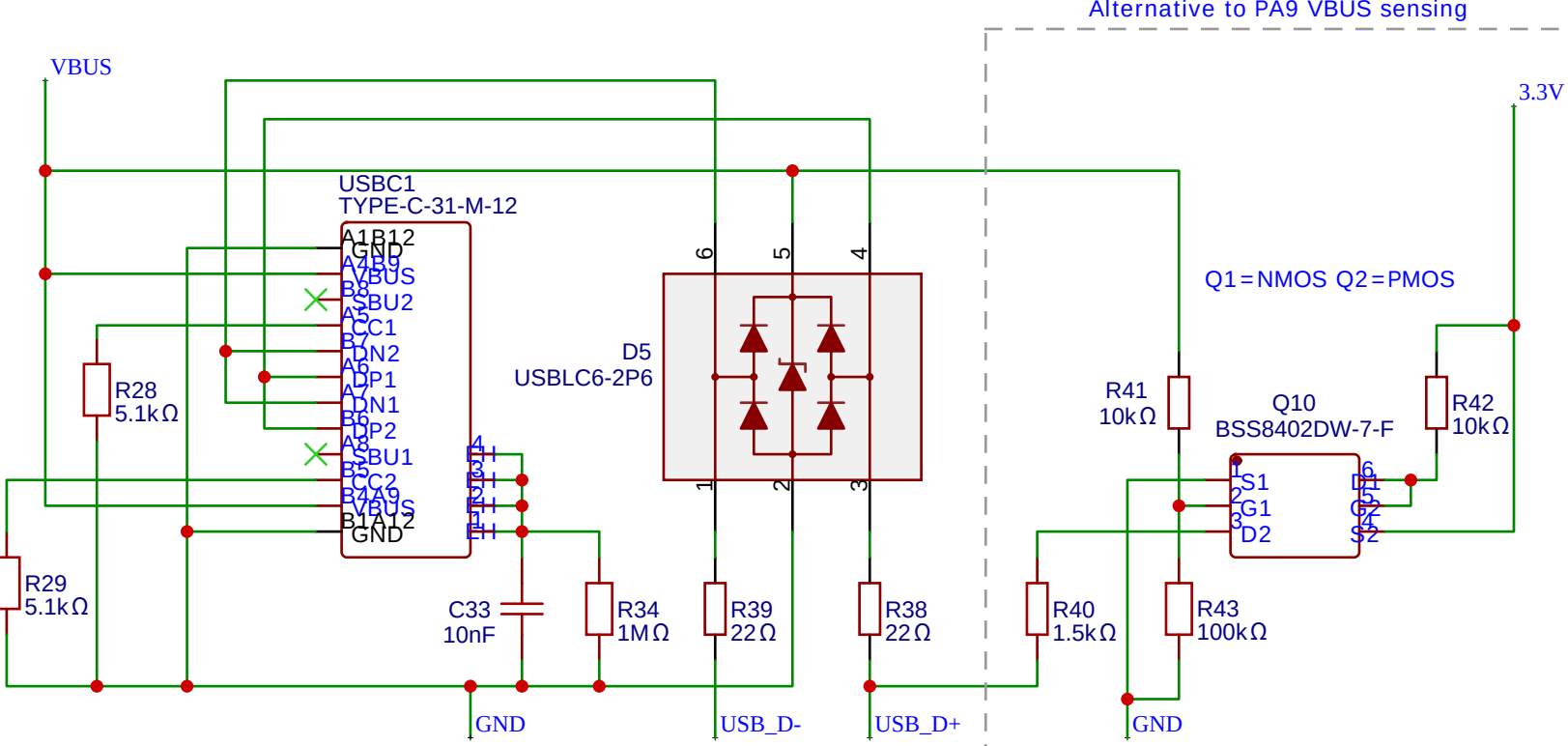
0.004R shunt = 1A/80mV - 40A/3200mV

Input filter R11, R12 and C19

Output filter R35/C16 for the MCU ADC



USB connector for programming (optional)



TITLE: DC Motor Controller		REV: 1.2
Company: KFCLabs		Sheet: 1/1
Date: 2026-07-28		Drawn By: sascha23095123423